

HTTP 5221

Critical Path

Web Development: Security & Usability

Summer 2026 (updated July 6, 2026)

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The Critical Path is the course schedule. It is an important document that lays out weekly and/or module objectives that you are responsible for to ensure academic success. Download and/or print out a copy of the Critical Path and refer to it regularly.

Week #	Date	Module	Topics	Activities	Type	Description	Weight	Due Date	Last Date to Submit
8	July 9	Level Up Your Code!	- object, class, constructor - querySelector, querySelectorAll, onclick, unload, keyup - variables scopes - window.location object (search, href, host, hostname, path, hash)	Lab Exercise 1	Individual	- Create a list of friends using JS. - Complete Quiz on knowledge of JS variables, objects. - Create a page using routing.	3%	July 11	July 15
9	July 16	Coding Standards	- B.E.M. naming convention - JS Lint "use strict"	Lab Exercise 2	Individual	- Create a form with Name, Salary and Postal code inputs JS validation. - Complete Quiz on knowledge of JS regex.	6%	July 18	July 22
10	July 23	Test Design	- Testing strategies, spec, requirements, bug prevention. - Dynamic, static testing, boundary conditions, tests to pass, to fail. - Unit testing, TDD	Lab Exercise 3	Individual	- Create a JS function to validate Humber ID. - Create unit tests for this function.	6%	July 25	July 29
11	Jul 30	Testing Frameworks & Encryption	- Encryption libraries with JS (md5, hash, salt) - Jasmine Testing Framework	Lab Exercise 4	Individual	- Create a JS function to guess a month. - Create Jasmine unit tests for this function.	4%	Aug 1	Aug 5

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				Assignment 1: Jasmine Testing	Individual	- Create a form with JS validation and error/success messages. - Create Jasmine unit tests for JS code.	15%	Aug 12	Aug 16
12	Aug 6	Hacker Prevention	- Hacking, black hat, white hat, grey hat - Intent & permission. - Penetration testing, exploits.	Assignment 2	Group	- Research a vulnerability of your choice. - Create html page with vulnerability description.	10%	Aug 8	Aug 12
13	Aug 13	End to End (E2E) Testing	- Test coverage, testing pyramid. - Integration testing, e2e, user testing, automated testing. - Cypress.	Lab Exercise 5	Individual	- Create a subscription form - Create Cypress tests for the form.	4%	Aug 15	Aug 19
14	Aug 20	Ad Hoc Testing	- Ad hoc testing, web testing checklist - functionality testing (valid html, css, js, links, forms, cookies). - Compatibility testing, performance testing, security testing, usability testing.	Lab Exercise 6	Individual	- Fix accessibility, CSS, HTML & performance issues - Use browser extensions for Ad Hoc testing.	2%	Aug 22	Aug 23
Final Marks due August 25									

Summary of Graded Components

Component	Module	Weight
Group Assignment	Security	10%
Individual Assignment	Security	15%
Lab Exercise (6)	Security	25%
Total		50%